

PdsA/DdPDE1 (DICDI, UniProt P12019) — Adjudication of two negative-regulation GO claims

Focus: function_assignment · **Hypothesis slug:** extracellular-pde-and-camp-pathway-regulation **Terms adjudicated:** GO:0110034 (negative regulation of adenylate cyclase-activating **glucose-activated** G protein-coupled receptor signaling pathway) and GO:0141162 (negative regulation of cAMP/PKA signal transduction).

Executive Judgment

Verdict: REFUTED / over-annotated for both terms *as assigned to PdsA*.

Both GO terms are already present on P12019 **only as IBA (phylogenetic, ECO:0000318) annotations** propagated from fungal cyclic-nucleotide phosphodiesterase orthologs via PANTHER family PTN002001416 (GO:0110034 ← *S. pombe* cgs2/Pde1, PomBase SPCC285.09c; GO:0141162 ← *C. albicans* PDE2, CGD CAL0000177603). Neither is backed by a *Dictyostelium* experiment.

- **Claim A (GO:0110034):** The GO term specifically denotes a **glucose-activated** GPCR→adenylate-cyclase pathway (the fungal Gpr1/Gpa2→Cyr1→cAMP→PKA glucose-sensing system). *Dictyostelium* has no such pathway; its adenylate-cyclase (ACA)-activating receptor cAR1 is **cAMP-activated, not glucose-activated**. PdsA's genuine negative-regulatory role — degrading the extracellular cAMP ligand of the cAR receptors — is real but is **the wrong biological pathway for this term**. The term is factually inapplicable, independent of PdsA being secreted.
- **Claim B (GO:0141162):** PdsA degrades the **extracellular** cAMP pool. PKA is activated by the **intracellular** cAMP pool, whose direct negative regulator is the cytosolic PDE **RegA** (Q23917). The direct catalytic step in cAMP/PKA signal transduction is RegA's, not PdsA's. PdsA can influence PKA only **indirectly** through cAR-receptor feedback/compartment coupling, which does not justify a direct BP annotation.

Most important caveat: This is a *compartment/term-specificity* judgement, not a claim that PdsA is irrelevant to cAMP signaling. PdsA is a bona-fide negative regulator of the *extracellular* cAMP relay/GPCR loop — that role is already captured by experimentally supported terms (GO:0007193 IMP, GO:1900115 IMP). The two seed terms should be removed as ortholog carry-over, not replaced with themselves.

Decisive primary evidence (added Iteration 2): Bader, Kortholt & Van Haastert 2007

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biochemically partition all seven Dictyostelium PDEs across three cyclic-nucleotide pools and state that **extracellular cAMP is degraded predominantly by DdPDE1 (= PdsA) and its homolog DdPDE7**, whereas **intracellular cAMP (which "mediates development") is degraded by DdPDE2 (= RegA) and DdPDE6**. This is a direct, quantitative pool assignment — not an inference from secretion — that separates PdsA (extracellular) from the RegA-controlled intracellular/PKA-relevant pool.

Correct-term note: GO does provide a non-glucose term, **GO:0106072 "negative regulation of adenylate cyclase-activating G protein-coupled receptor signaling pathway"**, and a cAMP-receptor pathway term **GO:0140582 "adenylate cyclase-activating G protein-coupled cAMP receptor signaling pathway"**. These, not the fungal glucose term GO:0110034, are the appropriate vocabulary if a receptor-pathway BP annotation is desired for PdsA's cAR/ACA context. (Note: GO:0043951 "negative regulation of cAMP-mediated signaling" is **obsolete** and unusable.)

Evidence Matrix

Citation	Evidence type	Stance	Claim tested	Key finding	Context
P 9405107	Mutant phenotype	Supports mechanism	PdsA controls extracellular cAMP	pdsA-null blocks coordinated chemotactic movement; PDE is secreted , controls extracellular cAMP	<i>D. discoideum</i> development
P 23473502	Imaging/direct	Supports mechanism	PdsA degrades extracellular cAMP	Extracellular PDE degradation sets cAMP near the relay threshold; dual +/- feedback	<i>D. discoideum</i> aggregation
P 36688866	Mutant phenotype	Qualifies	cAMP pool compartmentation	Adenylate cyclases make secreted cAMP for gene expression and intracellular cAMP for PKA activation ; PdsA is the extracellular PDE	Polysphondylium/Dictyostelium
P 30790701	Mutant phenotype	Refutes Claim B for PdsA	Neg. reg. of PKA = RegA	"the intracellular cAMP-specific phosphodiesterase RegA is a negative regulator of...PKA"	<i>D. discoideum</i>
P 11390363	Genetic/direct	Refutes Claim B for PdsA	Intracellular cAMP/PKA set by RegA	Intracellular cAMP & PKA activity controlled by synthesis and RegA degradation	<i>D. discoideum</i>
P 17040207	Direct assay (enzyme	Refutes Claim B; supports	Which PDE clears which cAMP pool	DdPDE1 (=PdsA) clears extracellular	<i>D. discoideum</i> , all 7 PDEs

Citation	Evidence type	Stance	Claim tested	Key finding	Context
	kinetics/pool partition)	correct term		cAMP; DdPDE2 (=RegA) clears intracellular cAMP that "mediates development"; basis of IMP GO:0007193 & GO:1900115	
QuickGO/UniProt P12019	Database/curation metadata	Refutes seed terms	Disputed terms are IBA	GO:0110034 & GO:0141162 are IBA (GO_REF:0000033) from fungal Pde1/Pde2 via PANTHER PTN002001416	cross-species
QuickGO GO:0110034 annotation set	Computational	Qualifies	Term is fungal glucose-signaling	34 annotations dominated by fungal cgs2/atf1/cgs1(PKA)/PDE1; Dictyostelium Pds enzymes added only by phylogeny	cross-species

Provenance artifacts (computed this run): `/tmp/go_decision_table.csv`, `/tmp/evidence_matrix.csv`, `/tmp/G00110034_composition.png`.

GO Curation Implications (leads — require curator verification)

GO term	Aspect	Current evidence on P12019	Lead action
GO:0110034 neg reg AC-activating glucose-activated GPCR signaling	BP	IBA only (from <i>S. pombe</i> cgs2/Pde1)	Remove / do not assign. Wrong pathway (glucose-sensing GPCR). Report as inappropriate IBA propagation.
GO:0141162 neg reg cAMP/PKA signal transduction	BP	IBA only (from <i>C. albicans</i> PDE2)	Remove from pdsA. Belongs to regA (Q23917), the intracellular cAMP/PKA regulator.
GO:1902660 neg reg glucose mediated signaling pathway	BP	IBA	Flag as non-core ; same fungal-ortholog family; candidate removal.
GO:0007193 adenylate cyclase-inhibiting GPCR signaling pathway	BP	IMP (P 17040207)	Retain — experimentally supported <i>Dictyostelium</i> role.
GO:1900115 extracellular regulation of signal transduction	BP	IMP (P 17040207)	Retain — correctly captures action on the extracellular signal pool (core BP).
GO:0004115 3',5'-cyclic-AMP phosphodiesterase activity	MF	IDA (dictyBase); EC 3.1.4.35/3.1.4.53	Retain — primary molecular function.
GO:0106072 neg reg adenylate cyclase-activating GPCR signaling (non-glucose)	BP	<i>not currently on P12019</i>	Candidate replacement for GO:0110034 if a receptor-loop BP is wanted; matches cAR/ACA (GO:0140582) context. Curator-verify against P 17040207 / 9405107.

Do **not** default to "protein binding": the informative MF (cAMP phosphodiesterase, GO:0004115) and the correct BP terms are already available.

Directionality nuance (curator-verify): the currently retained BP term GO:0007193 is *adenylate cyclase-inhibiting GPCR signaling*, but the characterized Dictyostelium biology is that cAR1 **activates** adenylate cyclase (ACA) and PdsA negatively regulates that loop by degrading the extracellular cAMP ligand. On directional grounds **GO:0106072** (*negative regulation of adenylate cyclase-activating GPCR signaling*) is a more accurate BP than GO:0007193. A curator may wish to consider replacing GO:0007193 with GO:0106072 (context GO:0140582, cAMP-receptor pathway), rather than simply retaining GO:0007193.

Mechanistic Scope

- **Direct molecular activity of PdsA:** hydrolysis of 3',5'-cyclic AMP (preferred) and cGMP to 5'-nucleotides (EC 3.1.4.35 / 3.1.4.53), acting on the **extracellular / cell-surface** cAMP pool (UniProt: "Secreted, extracellular space; Cell surface").
- **Direct cellular role:** clearing secreted cAMP so cAR-mediated relay can oscillate and cells stay responsive during aggregation

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This is genuine **negative regulation of a cAMP-receptor (GPCR) / adenylate-cyclase loop** — the concept behind Claim A but for a *cAMP*-activated, not *glucose*-activated, receptor.

- **What is NOT PdsA's direct step:** setting the intracellular cAMP concentration that binds PKA regulatory subunits. That is done by RegA (intracellular) and adenylate cyclases (synthesis)

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Any PdsA→PKA effect is a **downstream / feedback** consequence, not a direct transduction step.

Conflicts and Alternatives

- **Paralog/ortholog carry-over (primary conflict):** both seed terms are IBA from a PANTHER family that pools fungal intracellular glucose-signaling PDEs (*S. pombe* Pde1/cgs2, *C. albicans* PDE2) with *Dictyostelium* secreted PdsA. In fungi those PDEs really do negatively regulate glucose-activated GPCR→cAMP→PKA signaling; the terms are correct **there**, then over-propagated to PdsA whose biology (extracellular, cAMP-chemoattractant) differs.
- **Compartment confusion:** the seed hypothesis explicitly asks to separate the extracellular pool (PdsA) from the RegA-controlled intracellular pool — the evidence confirms these are the correct compartment assignments, and the two seed terms conflate them.
- **Term-label subtlety:** GO:0110034's "glucose-activated" wording could be mistaken for generic "adenylate cyclase-activating GPCR" signaling. Curators should note the qualifier makes it fungal-glucose-specific, so it is not a substitute for the cAMP-receptor role already covered by GO:0007193.

Knowledge Gaps

1. **Does PdsA measurably affect intracellular cAMP/PKA in vivo?** Checked: literature attributes PKA control to RegA; no assay shows PdsA setting intracellular cAMP. Matters because a genuine (if indirect) effect could justify a *regulates*-type annotation with an ISS/indirect qualifier — but current evidence favors indirect only. Resolve with: intracellular cAMP / PKA-reporter measurements in pdsA-null vs WT.
2. **Is there any Dictyostelium glucose-sensing GPCR→adenylate cyclase pathway?** Checked: not described; ACA-activating receptor is cAR1 (cAMP). Matters because GO:0110034 requires it. Resolve with: literature/annotation review confirming absence.
3. **RESOLVED (Iteration 2):**

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abstract read — it explicitly assigns extracellular-cAMP degradation to DdPDE1 (PdsA) and intracellular-cAMP degradation to DdPDE2 (RegA), confirming the compartment split. Remaining minor gap: the paper is enzyme-kinetics/pool-modeling, so the "IMP" evidence label on GO:0007193/GO:1900115 is curator interpretation; full-text confirmation of the receptor-signaling phenotype would further solidify the retained BP terms.

Discriminating Tests

- **pdsA-null vs regA-null, intracellular cAMP + PKA activity time-course:** regA-null elevates intracellular cAMP/PKA (precocious development); pdsA-null should not directly elevate the intracellular pool — cleanly separates Claim B ownership.
- **8Br-cAMP / PKA-agonist rescue:** rescues intracellular-cAMP/PKA deficits (as in acaA-acrA-double mutants,

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but is not expected to be the axis of pdsA phenotypes (which are extracellular-relay/chemotaxis).

- **Ortholog-context check:** confirm PANTHER PTN002001416 groups secreted Dictyostelium PDEs with intracellular fungal PDEs — evidence that the IBA propagation crosses a compartment/function boundary and should be blocked for PdsA.

Curation Leads (require curator verification)

- **Action:** Remove/withhold **GO:0110034** and **GO:0141162** from pdsA (P12019); both are IBA-only ortholog carry-overs crossing a compartment/pathway boundary. Consider flagging **GO:1902660** as non-core.
- **Optional replacement (curator-verify):** if a receptor-loop BP is desired, use the non-glucose **GO:0106072** (negative regulation of adenylate cyclase-activating GPCR signaling pathway), contextualized to the cAMP-receptor pathway **GO:0140582** — not the fungal glucose term GO:0110034. Note GO:0043951 is obsolete.
- **Reassign:** GO:0141162 is appropriate for **regA (Q23917)**, the intracellular cAMP/PKA negative regulator. Primary support:

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— "Intracellular cAMP is degraded by the DdPDE2 [RegA]..." vs "Extracellular cAMP is degraded predominantly by...DdPDE1 [PdsA]".

- **Retain:** GO:0004115 (MF, IDA), GO:0007193 and GO:1900115 (BP, IMP

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— these already capture PdsA's true negative-regulatory, extracellular role.

- **Candidate references / snippets to verify:**

- P 30790701

 — "the intracellular cAMP-specific phosphodiesterase RegA is a negative regulator of...PKA".
- P 11390363

 — "intracellular cAMP and PKA activity are controlled by...degradation by the cAMP-specific phosphodiesterase RegA".
- P 36688866

 — adenylate cyclases produce "secreted cAMP for...gene expression and intracellular cAMP for PKA activation".
- P 9405107

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- P 23473502

 — PdsA is the secreted extracellular-cAMP PDE governing relay.
- **Suggested questions for curator:** Should IBA annotations from PANTHER PTN002001416 be suppressed for the secreted Dictyostelium Pds clade? Is a "regulation of adenylate cyclase-activating (cAMP) GPCR signaling" term (non-glucose) already covered by GO:0007193 for this gene?
- **Suggested experiments:** intracellular cAMP/PKA-reporter assay in pdsA-null vs regA-null to formally exclude any direct PdsA role in the PKA-activating pool.

Limitations

Analysis used public UniProt/QuickGO metadata and PubMed/efetch abstracts; the fungal ortholog primary papers (*S. pombe* cgs2/Pde1; *C. albicans* PDE2) were not full-text read. Verdicts on IBA provenance are drawn from curation metadata (evidence code, GO_REF, with-from, PANTHER family PTN002001416) — high confidence — rather than re-derived phylogeny.

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was read at abstract level (Iteration 2) and directly documents the PdsA=extracellular / RegA=intracellular pool split.

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